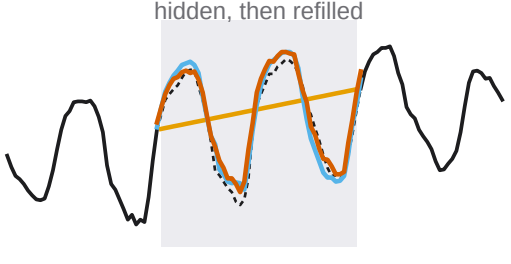
Weather records have gaps. Does filling them change the energy simulation?



We hid known hours in 17 complete weather-years, refilled them with 7 methods and simulated 6 buildings.

	length of the filled gap: 3–12 hours	2 days	1 week	
Annual heating	< 0.3%	< 0.4%	< 1.7%	barely
Annual cooling	< 1.1%	< 1.3%	< 5%	
Size of peak load	~1 W/m²	< 8 W/m²	< 17 W/m²	yes, for long gaps
Peak moved by > 1 day	23%	35%	51%	
Heatwaves lost	none	8%	20%	

Largest errors came from nearby stations that failed during extreme weeks.
Advice: for gaps longer than 1–2 days, switch to ERA5 reanalysis.